

## Multiple Choice

1. **Answer:** A

*Options:* A) salt; B) water; C) hydrogen gas; D) carbon monoxide

*Note:* Salt, or sodium chloride, is an ionic compound that forms a crystal lattice structure, where ions are held together by strong electrostatic forces, rather than existing as discrete molecules. This key characteristic distinguishes it from covalent compounds, which consist of individual molecules.

2. **Answer:** B

*Options:* A) The tile floor is colder than the carpet.; B) The tile floor conducts heat better than the carpet.; C) The carpet conducts heat better than the tile floor.; D) The tile floor is a better insulator than the carpet.

*Note:* The correct answer is based on the concept of thermal conductivity, which indicates that tile has a higher thermal conductivity than carpet. As a result, tile efficiently absorbs heat from our body, making it feel colder, whereas carpet retains heat, providing a warmer sensation.

3. **Answer:** B

*Options:* A) eye color.; B) athletic ability.; C) number of teeth.; D) shape of earlobes.

*Note:* Genes play a significant role in determining certain physical traits and predispositions, but athletic ability is influenced by a combination of genetic factors, environmental conditions, training, and personal motivation, making it impossible for genes alone to determine a person's athletic potential. Thus, while genetics can contribute to athletic performance, they do not exclusively dictate it.

4. **Answer:** A

*Options:* A) burns at a cooler temperature.; B) is always larger than other suns.; C) has greater density and gravity.; D) fuses hydrogen to make helium.

*Note:* A red giant star differs from main sequence stars because it has exhausted the hydrogen fuel in its core and is now fusing helium or heavier elements, which occurs at a cooler temperature compared to the hydrogen fusion that powers main sequence stars. This lower temperature causes the star to expand and take on a reddish hue, distinguishing it from the hotter, more luminous main sequence stars.

5. **Answer:** D

*Options:* A) pollution; B) erosion; C) water tables rise; D) lack of precipitation

*Note:* A natural cause of rivers drying up is a lack of precipitation because rivers rely on rainfall and snowmelt to replenish their water levels; when these sources of water decrease, the flow of the river diminishes, leading to dry conditions.

## True / False

6. **Answer:** True

*Options:* A) True; B) False

*Note:* The outer layers of astronauts' space suits are designed to be reflective in order to shield them from the intense sunlight and harmful radiation in space, helping to regulate their temperature and protect their skin from ultraviolet rays.

7. **Answer:** True

*Options:* A) True; B) False

*Note:* Landslides indeed represent a sudden change to Earth's surface, as they occur when gravity causes loose rocks and soil to rapidly move downhill, often triggered by factors such as heavy rainfall, earthquakes, or human activity. This process alters the landscape, making the statement true.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* The statement is false because while sulfur dioxide can originate from natural sources such as volcanoes, human activities, particularly the burning of fossil fuels and industrial processes, contribute significantly to sulfur dioxide emissions, allowing for substantial human control over its levels in the atmosphere.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* Rubbing a coin with cold hands will only transfer a small amount of heat away from the coin, while placing it in cool water allows for a more efficient heat exchange, making the coin colder than simply rubbing it with hands.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* Incomplete metamorphosis is characterized by the life cycle of certain insects, such as grasshoppers and cockroaches, where the organism develops through a series of nymph stages that resemble the adult form, bypassing a distinct pupal phase.

## Long Form Response

11. **Answer:** The alternative energy resource that most closely resembles coal in its usage is biomass, particularly in the form of wood pellets. Biomass can be burned in power plants similar to coal, providing a renewable energy source that helps reduce greenhouse gas emissions. However, challenges include the need for sustainable sourcing to prevent deforestation and the potential for air pollution if not managed properly, making it crucial to balance its benefits with these environmental concerns.

*Note:* The answer correctly identifies biomass, particularly wood pellets, as an alternative energy resource that can be used similarly to coal in power generation, emphasizing its potential for reducing greenhouse gas emissions while also acknowledging the challenges of sustainable sourcing and air pollution management. This balanced

perspective illustrates the complexities of transitioning to renewable energy sources in comparison to traditional fossil fuels.

12. **Answer:** Organisms classified as Eukarya have complex cells with a defined nucleus and membrane-bound organelles, allowing for specialized functions, while Archaea and Bacteria are prokaryotic, meaning they lack a nucleus and have simpler structures. Eukarya includes plants, animals, and fungi, which play diverse ecological roles such as producers and consumers, whereas Archaea and Bacteria are crucial for nutrient cycling, with some capable of extreme survival conditions. These differences highlight the complexity of Eukarya compared to the more ancient and diverse forms of life found in Archaea and Bacteria.

*Note:* The answer correctly identifies that Eukarya organisms possess complex cellular structures with a nucleus and organelles, enabling specialized functions, while Archaea and Bacteria are simpler, prokaryotic organisms that play essential roles in nutrient cycling and can thrive in extreme environments, thus highlighting their ecological significance.